Transport Graph Lab Demo

Coherent and Dissipative Transport in Simple Open Quantum Networks

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Abstract

This short report documents the numerical experiments currently implemented in the transport-graph laboratory. The goal is to compare simple graph topologies under the same open-system model, to explain what the sink represents, and to show how the main observable (sink efficiency) changes when the dephasing rate is varied. The baseline comparison includes a complete graph, a ring, and a chain, with population dynamics and coherence used as secondary diagnostics.

1 What these experiments are

The present numerical experiments are not laboratory measurements. They are controlled open-system simulations designed to answer a clean physics question: how graph topology changes transport toward a target channel when coherent hopping competes with decoherence.

Each simulated experiment follows the same logic:

1. choose a graph topology;
2. prepare one initial excitation on a selected site;
3. evolve the state with coherent hopping plus dissipative channels;
4. measure how much population reaches the sink;
5. compare the result across dephasing rates and topologies.

2 Physical model and references

The network is described in the single-excitation sector by the tight-binding Hamiltonian

$$H = \sum_{i=1}^{N} \epsilon_i |i\rangle\langle i| + \sum_{i \neq j} J_{ij} |i\rangle\langle j|. \tag{1}$$

Open-system evolution is modeled with the Lindblad master equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right), \tag{2}$$

which is the standard minimal framework for a first study of coherent and dissipative transport in small open systems [1, 2].

The transport-specific motivation comes from the literature on dephasing-assisted or environment-assisted transport in quantum networks [3–9]. The optional long-term bridge to experiment is motivated by programmable quantum platforms such as superconducting circuits [10–12].

Table 1: Automatically generated summary of the current baseline graph comparison.

Scenario	Best regime	η_{coh}	η_{best}	$\gamma_{\phi}^{\mathrm{best}}$
Complete graph	intermediate	0.300	0.798	3.200
Ring graph	coherent	0.795	0.795	0.000
Chain graph	coherent	0.812	0.812	0.000

3 What the sink means

The sink is an absorbing target state that represents successful transport. It is not an additional graph node. Instead, it is the bookkeeping channel that receives population from the chosen trap site. The final sink population is therefore the transport-success metric,

$$\eta(T) = \rho_{ss}(T). \tag{3}$$

This construction is useful because it separates:

- coherent motion inside the graph;
- successful capture into the target channel;
- unwanted dissipation into a distinct loss channel.

4 Main and secondary observables

The main observable is the sink efficiency:

$$\eta(T) = \rho_{ss}(T). \tag{4}$$

The secondary observables are:

$$P_i(t) = \rho_{ii}(t), \tag{5}$$

which track population flow across the graph, and

$$C_{\ell_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|, \tag{6}$$

which provides a direct coherence diagnostic in the site basis.

5 Baseline topologies

The current baseline uses three families:

- a complete graph, which is useful for exposing interference-assisted and interference-blocked pathways;
- a ring, which allows coherent circulation and sensitivity to disorder;
- a chain, which gives the simplest directional transport geometry.

6 Current simulation results

The following table is generated automatically from the latest run of the graph lab baseline.

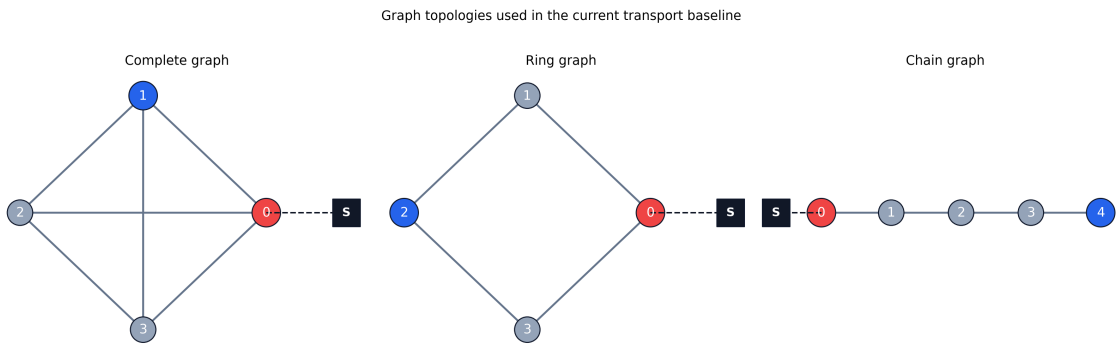

Figure 1: Topology overview of the graphs used in the current baseline. The colored nodes indicate the initial site and the trap site, while the square node S represents the sink channel used to measure successful transport.

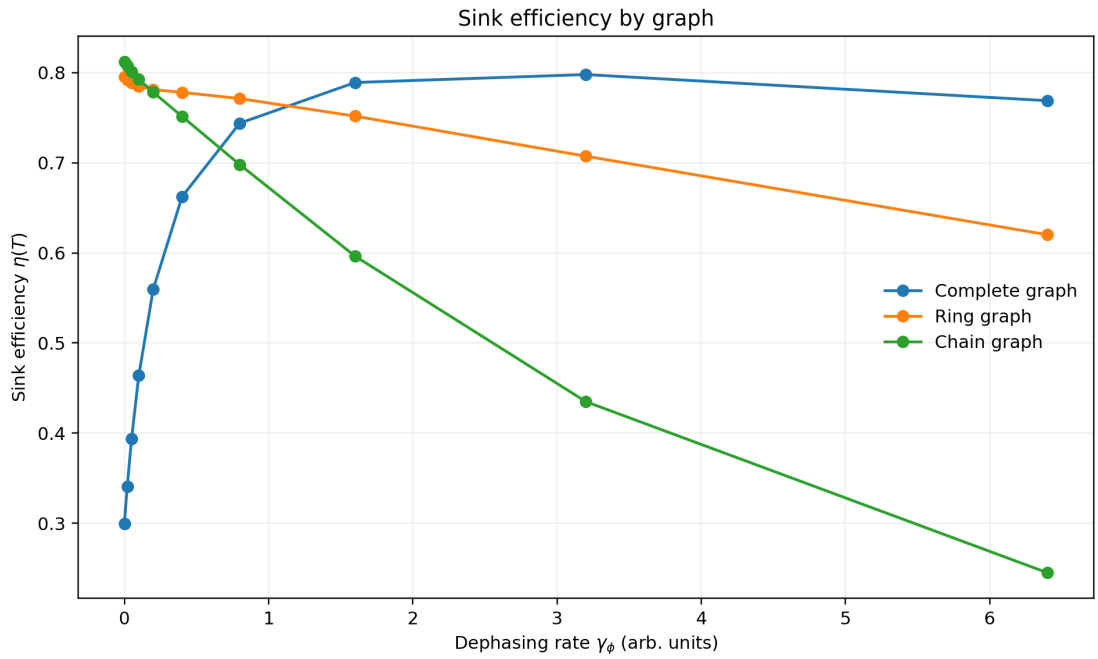

Figure 2: Main result: sink efficiency as a function of the dephasing rate for the baseline graph family comparison.

7 How to change the simulations

The editable configuration file is:

`configs/transport_graph_lab_config.json`

The main parameters you can change are:

- `seed`;
- `n_sites`;
- `topology`;
- `initial_site`;
- `trap_site`;

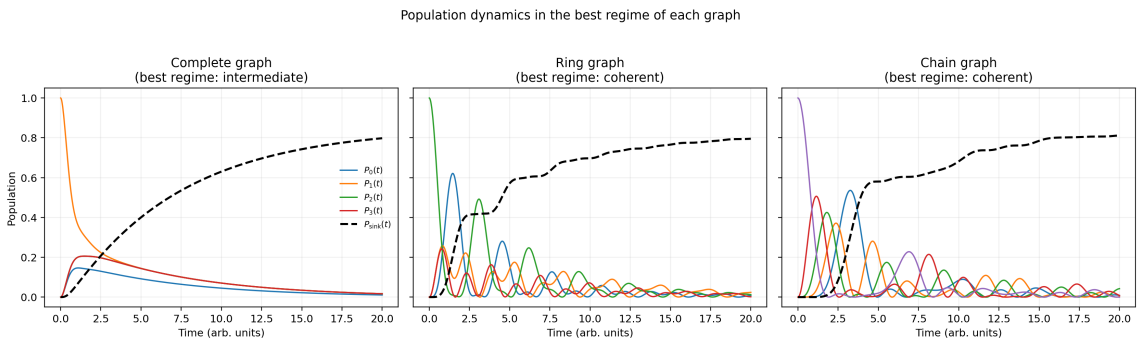

Figure 3: Population dynamics in the best regime identified for each graph.

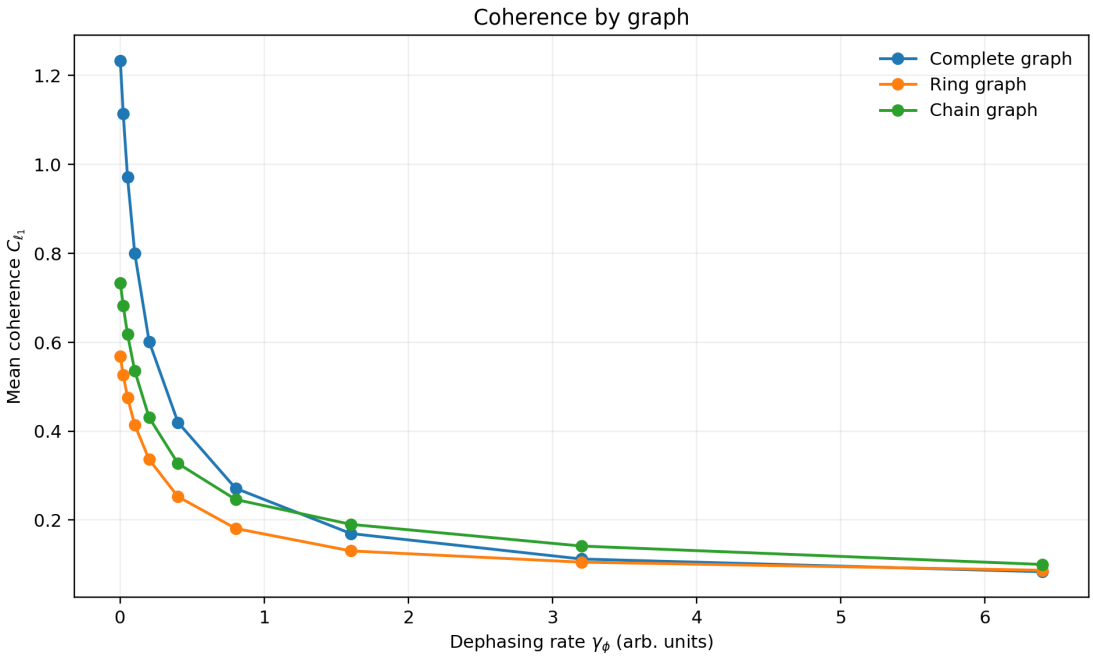

Figure 4: Coherence diagnostic as a function of the dephasing rate for the same graph family comparison.

- `coupling_hz`;
- `sink_rate_hz`;
- `loss_rate_hz`;
- `disorder_strength_hz`;
- `dephasing_rates_hz`.

After editing the configuration, the simulation is regenerated by running:

```
python scripts\run_transport_graph_lab.py
```

The lab also generates one GIF animation per scenario, showing how the node populations, sink occupation, and loss evolve in the best regime identified for that graph.

8 Interpretation of the current baseline

The present baseline already shows three qualitatively different behaviors. For the complete graph, the best regime is intermediate, which means that moderate dephasing improves sink

efficiency relative to the coherent limit. For the ring and the chain, the best performance remains at the coherent limit, while strong dephasing suppresses transport. This is already enough to justify the central research direction of the project: topology and decoherence cannot be discussed separately.

References

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